A Recipe for Long-Context Reasoning in Large Language Models via On-Policy Optimization and Distillation

Miguel Moura Ramos[1,2] **Duarte M. Alves**[1,2] **André F. T. Martins**[1,2,3,4]
[1]Instituto Superior Técnico, Universidade de Lisboa
[2]Instituto de Telecomunicações [3]TransPerfect [4]ELLIS Unit Lisbon

Abstract

Existing approaches to post-train models for long-context tasks face complementary limitations: (i) supervised fine-tuning (SFT) provides stable supervision but suffers from exposure bias; (ii) reinforcement learning methods such as Group Relative Policy Optimization (GRPO) train on model-generated trajectories but struggle with long-horizon credit assignment and sparse rewards; and (iii) on-policy distillation (OPD) provides dense token-level guidance but does not directly optimize task rewards. We study these complementary strategies for long-context alignment and derive a recipe that combines GRPO with OPD-style teacher guidance: the student learns from its own rollouts using outcome-level rewards, while a stronger teacher provides dense token-level regularization in place of the standard reference policy. This is especially useful when process-level supervision is difficult to obtain. To support this study, we introduce LongBlocks, a synthetic multilingual dataset spanning multi-hop reasoning, contextual grounding, and long-form generation. Through controlled ablations, we isolate the roles of cold-start initialization, teacher anchoring, and data mixing, showing that our recipe yields a more stable and effective path to long-context reasoning than GRPO or OPD while preserving short-context capabilities.[1]

1 Introduction

Real-world applications increasingly require large language models (LLMs) to reason over and integrate information from extremely long contexts. This arises in tasks requiring understanding large codebases (Jimenez et al., 2024), synthesizing information across collections of documents (Yang et al., 2018), and maintaining coherence across multi-session interactions spanning hundreds of thousands of tokens (Mei et al., 2025; Liu et al., 2024). Yet, aligning LLMs for long-context reasoning remains challenging due to multiple factors, including positional extrapolation, attention dilution, and exposure bias (Peng et al., 2024; Liu et al., 2024; Bengio et al., 2015). These issues make long-context post-training qualitatively different from simply extending a short-context alignment recipe to longer sequences. Existing post-training strategies lack a standardized, open recipe that reliably delivers strong long-context performance (OpenAI, 2025; Yang et al., 2025; Comanici et al., 2025). Current efforts are further constrained by limited data and compute: off-policy alignment datasets are typically small and focused on relatively simple retrieval and summarization tasks, which restricts the diversity of long-context supervision (Bai et al., 2024a; Chen et al., 2024). At the same time, long-context settings remain relatively underexplored for on-policy alignment (Ouyang et al., 2022) and distillation (Agarwal et al., 2024). Existing long-context work has largely focused on preference optimization (Bai et al., 2024c; Zhang et al., 2025) or on-policy methods with limited sampling (Wan et al., 2025), leaving the interaction between outcome-level rewards and dense teacher guidance poorly understood in long-context regimes.

Motivated by these gaps, we ask: **how can post-training components be integrated for efficient, stable, and generalizable long-context reasoning?** We propose a two-stage recipe: an off-policy warm-up stage followed by teacher-anchored on-policy RL, where outcome-reward optimization on student-generated rollouts is performed jointly with OPD-style teacher guidance. Warm-up improves rollout quality before RL, reward optimization adapts the model under its own trajectory distribution, and teacher guidance replaces the standard reference-policy anchor with dense token-level regularization from a stronger model. This teacher signal provides intermediate supervision when process-level rewards are difficult to obtain,

[1]All resources are available on Hugging Face.

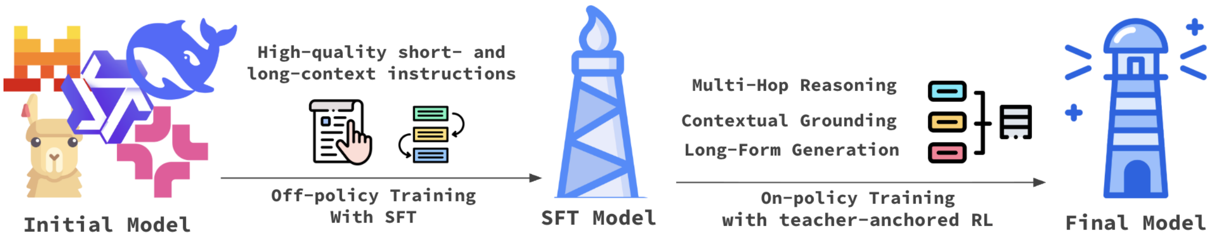

Figure 1: Illustration of our post-training recipe for building a long-context reasoning model.

adapting GRPO to the long-context regime where sparse rewards, long rollouts, and short-context retention must be balanced.

To evaluate this recipe, we conduct a controlled empirical comparison of long-context alignment strategies in a Qwen3-based setup (Figure 1). We isolate the roles of off-policy warm-up, outcome-reward GRPO, reference-policy anchoring, on-policy distillation, and teacher-anchored reward optimization under matched training budgets. This allows us to measure how dense teacher guidance interacts with sparse rewards, long rollouts, and short-context retention. To facilitate research on long-context post-training, we also introduce **LONG-BLOCKS**, a synthetic multilingual long-context dataset comprising multi-hop reasoning, contextual grounding, and long-form generation examples. Together, the recipe and dataset substantially improve long-context performance in a 1.7B student model without sacrificing its short-context capabilities.

2 Preliminary

2.1 Language Generation as a MDP

We formulate autoregressive language generation as a token-level Markov Decision Process (MDP) $\mathcal{M} = (S, A, r, T)$, where S is the set of token-prefix states, $A = \mathcal{V}$ is the action space given by the model vocabulary, r is the reward function, and T is the transition function. At each generation step t, the state $s_t \in S$ consists of the prompt p concatenated with all previously generated tokens: $s_t = [p_1, \ldots, p_M, o_1, \ldots, o_{t-1}]$. The action at time t is the next token $o_t \in A$. The transition function is deterministic: after action o_t, the next state is formed by appending the token, $s_{t+1} = T(s_t, o_t) = [s_t; o_t]$. The initial state s_1 is induced by the prompt $p \sim \mathcal{P}$, where $\mathcal{P}$ denotes the distribution over prompts. An episode ends when the policy outputs an end-of-sequence token [eos] or when the total token budget is reached. A policy is an autoregressive conditional distribution $\pi_\theta(\cdot|s_t)$, and the reward for a trajectory $\tau = (p, o)$ is the sum of per-step rewards: $R(p, o) = \sum_{t=1}^{|o|} r(s_t, o_t)$. We consider the undiscounted finite-horizon setting, which is standard for sequence generation tasks.

The agent typically maximizes a KL-regularized expected return (Schulman et al., 2018):

$$\mathcal{J}(\pi_\theta) = \mathbb{E}_{p\sim\mathcal{P},\, o\sim\pi_\theta(\cdot|p)} \Bigg[R(p, o) - \beta_{\text{ref}} \sum_{t=1}^{|o|} D_{KL}\big(\pi_\theta(\cdot|s_t) \,\|\, \pi_{\text{ref}}(\cdot|s_t)\big) \Bigg]. \quad (1)$$

Here π_{ref} is a reference policy, and β_{ref} is a KL regularization weight used in RLHF to prevent the tuned policy from drifting into regions where the learned reward model is unreliable. The KL term can be estimated on sampled trajectories using per-token log-ratios, $\log \pi_\theta(o_t|s_t) - \log \pi_{\text{ref}}(o_t|s_t)$. Following prior work (Liu et al., 2025; Yu et al., 2025; Olmo et al., 2025; Team et al., 2025), we set $\beta_{\text{ref}} = 0$, eliminating the need for a reference policy and reducing memory and compute costs.

2.2 Off-Policy Training

Off-policy supervision, primarily via supervised fine-tuning (SFT), provides a stable and data-efficient approach to bootstrap long-context behavior and can be viewed as a form of imitation learning (Ross et al., 2011) for language models. Formally, given an expert dataset $\mathcal{D}$ of input–output pairs (x, y), we optimize the model parameters θ by minimizing the token-level negative log-likelihood:

$$\mathcal{L}_{\text{SFT}}(\theta) = \mathbb{E}_{(x,y)\sim\mathcal{D}} \left[-\sum_{t=1}^{|y|} \log \pi_\theta(y_t \mid x, y_{<t}) \right]. \quad (2)$$

This objective provides dense token-level supervision, making it effective for instilling task adherence (Raffel et al., 2020; Ouyang et al., 2022).

When a strong teacher model is available, knowledge distillation (Hinton et al., 2015) can improve training by aligning the student policy π_θ with the teacher policy π_{T}. This is typically done by minimizing the token-level Kullback–Leibler divergence between the two distributions:

$$\mathcal{L}_{\mathrm{KD}}(\theta) = \mathbb{E}_{(x,y)\sim\mathcal{D}}\Big[\sum_{t=1}^{|y|} D_{KL}\big(\pi_{\mathrm{T}}(\cdot \mid x, y_{<t}) \,\|\, \pi_\theta(\cdot \mid x, y_{<t})\big)\Big]. \quad (3)$$

This distillation objective encourages the student to match the teacher's token distributions at each prefix, providing fine-grained guidance that can improve generalization beyond standard SFT. However, because both SFT and distillation rely on static expert trajectories, they remain off-policy and suffer from exposure bias (Ranzato et al., 2016; Bengio et al., 2015): they optimize only on data-distribution prefixes and do not train recovery from the model's own off-distribution states. This can degrade generation quality by amplifying errors during inference (Arora et al., 2022; Zhang et al., 2019; Chiang and Chen, 2021), especially in long-context settings, where retrieval distractors, positional extrapolation, length calibration, and optimization instability can compound over long rollouts.

2.3 On-Policy Training

On-policy RL approaches, such as Group Relative Policy Optimization (GRPO) (Shao et al., 2024), address an important limitation of off-policy training approaches by conditioning updates on the model's own rollouts. GRPO avoids exposure bias by optimizing the policy under the distribution induced by its own rollouts, enabling adaptation to long-range dependencies. Unlike standard PPO (Schulman et al., 2017), GRPO simplifies value estimation by removing the need for a separate value function critic. Instead, it estimates the baseline from a group of sampled completions $\{o_1, o_2, \ldots, o_G\}$ for a given prompt p. The objective function, with $\beta_{\mathrm{ref}} = 0$, can be defined as:

$$\mathcal{J}_{\mathrm{GRPO}}(\theta) = \mathbb{E}_{p\sim\mathcal{P},\, \{o_i\}_{i=1}^G \sim \pi_{\theta_{\mathrm{old}}}(\cdot|p)}\Bigg[\frac{1}{G}\sum_{i=1}^{G} \frac{1}{|o_i|}\sum_{t=1}^{|o_i|} \min\Big(\rho_{i,t}(\theta)\,\hat{A}_{i,t},\; \bar{\rho}_{i,t}(\theta)\,\hat{A}_{i,t}\Big)\Bigg], \quad (4)$$

where

$$\rho_{i,t}(\theta) = \frac{\pi_\theta(o_{i,t} \mid p, o_{i,<t})}{\pi_{\theta_{\mathrm{old}}}(o_{i,t} \mid p, o_{i,<t})},$$

$$\bar{\rho}_{i,t}(\theta) = \mathrm{clip}(\rho_{i,t}(\theta),\, 1-\epsilon,\, 1+\epsilon)\,,$$

and

$$\hat{A}_{i,t} = \frac{R(p, o_i) - \mathrm{mean}(\{R(p, o_1), \ldots, R(p, o_G)\})}{\mathrm{std}(\{R(p, o_1), \ldots, R(p, o_G)\})}.$$

Standalone GRPO is attractive for long-context alignment because it trains on model-generated trajectories, but its sparse rewards make optimization sample-inefficient and high-variance. It also remains susceptible to reward hacking (Liu et al., 2025; Yu et al., 2025), especially during long-context training where trajectory-level rewards are delayed and the lack of fine-grained supervision can hurt stability. An alternative approach that provides denser supervision is on-policy distillation (OPD), which guides the student policy toward a strong teacher by minimizing the reverse token-level KL divergence on student-generated trajectories, thereby providing token-level guidance while remaining on-policy (Agarwal et al., 2024; Yang et al., 2025; Lu and Lab, 2025). Unlike regularization toward a fixed reference policy, OPD aligns the student to the teacher on prefixes induced by student rollouts. Using reverse KL makes the distillation objective student-centered, focusing supervision on trajectories the student visits, which makes it a natural fit for on-policy RL. We define the OPD objective as

$$\mathcal{L}_{\mathrm{OPD}}(\theta) = \mathbb{E}_{p\sim\mathcal{P},\, o\sim\pi_\theta(\cdot|p)}\Bigg[\frac{1}{|o|}\sum_{t=1}^{|o|} D_{KL}\big(\pi_\theta(\cdot \mid p, o_{<t}) \,\|\, \pi_{\mathrm{T}}(\cdot \mid p, o_{<t})\big)\Bigg]. \quad (5)$$

However, OPD alone does not perform reward optimization and, therefore, cannot fully exploit reward-based feedback. In Section 3, we build on this observation by pairing GRPO-style outcome optimization with an on-policy teacher anchor, yielding a teacher-anchored reward-optimization objective for long-context alignment.

3 Recipe for Long-Context Reasoning

The preceding discussion suggests that no single ingredient is sufficient: off-policy supervision is stable but static, sparse-reward RL is adaptive but unstable, and OPD provides dense guidance but does not optimize task rewards. As illustrated in Figure 1, we therefore use a unified training pipeline that combines reward optimization with dense teacher guidance. This enables more stable policy improvement at extended sequence lengths while preserving short-context performance.

Cold-start Stage: Off-policy warm-up for stable long-horizon rollouts. The pipeline begins with SFT warm-up, which provides dense token-level supervision, improves long-context rollout coherence, and stabilizes subsequent on-policy learning. This is especially important in long-horizon settings, where low-quality or high-variance rollouts can make advantage estimation and KL-regularized updates brittle (Stiennon et al., 2020; Ouyang et al., 2022; Bai et al., 2022). We validate this stage in §4.3, including comparisons between SFT- and KD-based initialization.

RL Stage: Teacher-anchored on-policy reward optimization. After initialization, we perform on-policy RL with GRPO (Eq. 4) to optimize long-horizon task rewards under the model's own rollout distribution. In long-context settings, however, optimizing sparse outcome rewards alone can be unstable, degrading reasoning structure and short-context skills. We therefore adapt KL-regularized GRPO by replacing the standard reference-policy anchor with a stronger teacher. This preserves the outcome-reward objective while adding OPD-style dense token-level guidance on student-generated trajectories. Given a prompt $p \sim \mathcal{P}$, we sample a group of G rollouts $\{o_i\}_{i=1}^{G} \sim \pi_{\theta_{\text{old}}}(\cdot \mid p)$ and compute normalized advantages $\hat{A}_{i,t}$. Using the importance ratios $\rho_{i,t}(\theta)$ and clipped ratios $\bar{\rho}_{i,t}(\theta)$ from Eq. 4, we maximize

$$\mathcal{J}_{\text{ours}}(\theta) = \mathbb{E}_{p\sim\mathcal{P},\, \{o_i\}_{i=1}^{G}\sim\pi_{\theta_{\text{old}}}(\cdot|p)} \left[\frac{1}{G}\sum_{i=1}^{G} \frac{1}{|o_i|}\sum_{t=1}^{|o_i|} \left(\min\left(\rho_{i,t}(\theta)\,\hat{A}_{i,t},\ \bar{\rho}_{i,t}(\theta)\,\hat{A}_{i,t}\right) - \beta\, D_{\text{KL}}(\pi_\theta(\cdot \mid p, o_{i,<t}) \,\|\, \pi_{\text{T}}(\cdot \mid p, o_{i,<t})) \right) \right], \tag{6}$$

where $\beta \geq 0$ controls the strength of teacher regularization. The first term is the GRPO clipped surrogate for optimizing outcome rewards, while the teacher KL term anchors the student to the teacher along its sampled trajectories, providing dense intermediate guidance.

LongBlocks: A Multilingual Dataset to Tailor LLMs for Long-Context Reasoning. To support long-context training, we introduce LongBlocks, a multilingual synthetic dataset used for both off-policy warm-up and on-policy alignment. It contains 193,219 question–answer pairs generated from the raw document sources summarized in Table 1, covering more than 30 natural languages and 15 programming languages. We construct LongBlocks with a two-stage pipeline: first, we sample long, self-contained documents and prompt `Qwen3-Next-80B-A3B-Thinking` with the templates in Appendix A to generate question–answer pairs requiring evidence integration across extended contexts; second, we apply fuzzy deduplication and LLM-as-a-judge verification to remove malformed, weakly grounded, or locally answerable examples. Unlike prior long-context instruction datasets that primarily target document QA or instruction following, LongBlocks is designed to provide reasoning-oriented long-context supervision, with tasks and answers that encourage evidence aggregation, multi-hop reasoning, summarization, classification, and grounded long-form generation. We allocate examples uniformly across these task families at generation time, so that the mixture is not dominated by any single supervision format. Appendix A compares LongBlocks with existing long-context instruction datasets and details its benchmark-overlap controls and task-family balancing. To mitigate catastrophic forgetting across context lengths, we maintain a fixed 10/90 token-level mixture of short- and long-context data during both SFT and RL, drawing short-context samples from `Nemotron-Post-Training-Dataset-v2` and long-context samples from LongBlocks. We analyze this mixture in §4.3, showing that the 10/90 ratio is sufficient to preserve short-context performance, whereas larger short-context shares increasingly dilute long-context gains.

Training details. We initialize from `Qwen3-1.7B` and apply the two-stage pipeline in §3. We extend the context length to 128K without changing the original positional parameterization, setting the RoPE base frequency to 1×10^6 (Su et al., 2024). For SFT, we train for two epochs on the data mixture above with learning rate 1×10^{-5} and a batch size of four million tokens. We optimize target-token cross-entropy with AdamW (Kingma and Ba, 2015), weight decay 0.01, `bfloat16` mixed

Source	Coverage	Samples
Institutional Books (Cargnelutti et al., 2025)	3 natural languages (de, en, fr)	**107,908**
FineWeb2-HQ (Messmer et al., 2025)	9 natural languages (de, el, es, fr, hu, it, nl, sv, zh)	**39,113**
ArXiv (Clement et al., 2019)	English	**13,969**
Project Gutenberg Books (Gutenberg)	10 natural languages (de, en, fr, hu, it, nl, pl, pt, sv, zh)	**11,009**
Wikipedia (Foundation)	30 natural languages (ar, ca, da, de, el, es, . . . , uk, zh)	**10,718**
Stack-Edu (Allal et al., 2025)	15 programming languages (Python, C++, . . . , Markdown)	**5,430**
StackExchange (Kandpal et al., 2025)	English	**5,072**
Total		**193,219**

Table 1: Multilingual sources used for LONGBLOCKS. Appendix A details the language-wise composition.

precision, sequence packing (Raffel et al., 2020), and document masking to prevent cross-document attention (Grattafiori et al., 2024). For RL, we optimize Eq. 6 on the same mixture. Following Olmo et al. (2025), we use verifiable rewards for short-context tasks: SymPy for math, unit tests for coding, and binary constraint satisfaction for instruction following. For chat and long-context tasks, we use `Qwen3-4B` as a reference-conditioned LM judge that assigns binary rewards by comparing responses to references, following Lee et al. (2024). Appendix A reports judge-agreement analysis. We use outcome-level rewards because many tasks lack canonical step-by-step decompositions. The teacher is `Qwen3-32B`, with $\beta = 0.5$; we analyze sensitivity in §4.3. RL uses Adam with learning rate 1×10^{-6}, $(\beta_1, \beta_2) = (0.9, 0.95)$, gradient clipping 1.0, constant learning rate, clipping threshold 0.2, and 8 rollouts per prompt. Training used 4 NVIDIA H200 GPUs. All RL comparisons use a one-epoch budget and identical rollout settings.

4 Experiments

4.1 Experimental Setup

We compare our final model against the base model, `Qwen3-1.7B`, the SFT model from the cold-start stage, and strong external long-context baselines.

Decoding configuration. All models are evaluated with `lm-eval-harness` using a common decoding configuration: temperature 0.6, top-p = 0.95, top-k = 20, presence penalty 1.5, and a maximum generation length of 32,768 tokens, following Yang et al. (2025). To reduce variance from stochastic decoding, we run each benchmark 5 times and report the mean, with standard deviations provided in Appendix B, as in Olmo et al. (2025).

Short-context evaluation. We evaluate on IFEval (Zhou et al., 2023), IFBench (Pyatkin et al., 2025), MMLU-Pro (Wang et al., 2024), GPQA ♦ (Rein et al., 2024), GSM8K (Cobbe et al., 2021), MATH-500 (Lightman et al., 2024), HumanEval (Chen et al., 2021), HumanEval+ (Liu et al., 2023), MMMLU (OpenAI, 2024), and MGSM (Shi et al., 2022), covering instruction following, reasoning, multilinguality, and coding.

Long-context evaluation. We evaluate at 128K tokens on RULER (Hsieh et al., 2024), OneRuler (Kim et al., 2025), LongBench (Bai et al., 2024b), and ∞Bench (Zhang et al., 2024), covering retrieval, diagnostic tasks, multilingual aggregation, and long-context reasoning.

4.2 Main Results and Findings

Our recipe improves long-context performance without degrading short-context tasks. Figure 2 and Table 2 show that our recipe substantially improves long-context performance while preserving short-context accuracy. Short-context benchmarks remain stable or improve slightly, with the largest gains in code generation. On long-context benchmarks, improvements are consistent: average RULER performance rises from 64.5 to 74.8, with larger gains at longer sequence lengths, especially at 64K and 128K tokens (11.0→44.5 at 128K). Similar trends on ∞Bench and LongBench suggest that the gains are not limited to a single diagnostic benchmark. Appendix B provides per-task and multilingual results, as well as run-to-run uncertainty estimates.

Model	RULER						
	Avg.	4K	8K	16K	32K	64K	128K
Qwen3-1.7B	64.5	87.4	84.2	81.0	72.0	51.5	11.0
Ours	**74.8**	**88.3**	**85.4**	**84.7**	**80.4**	**65.6**	**44.5**

Table 2: RULER performance across sequence lengths.

Our recipe shifts the short-/long-context performance frontier. Figure 3 shows that our recipe

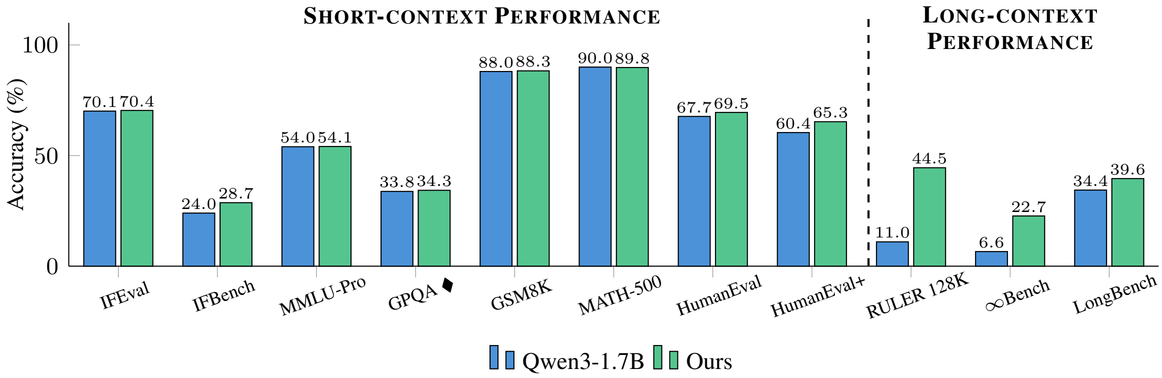

Figure 2: Short- and long-context benchmark performance for the base model and the final recipe.

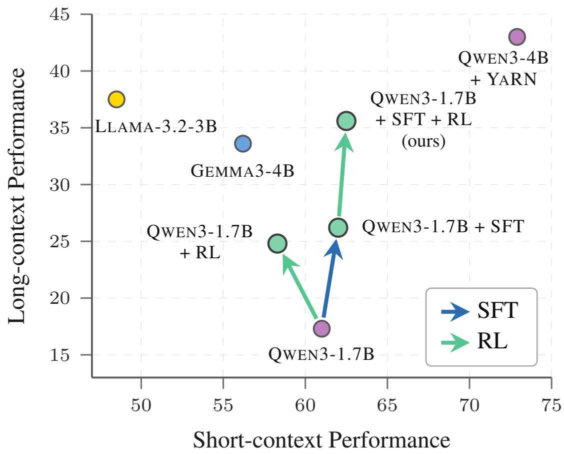

Figure 3: Short-/long-context performance frontier.

outperforms both the base model and stronger external long-context baselines without sacrificing short-context performance. The SFT-only checkpoint already improves long-context ability while preserving short-context competence, and teacher-anchored RL adds further gains. By contrast, applying the same RL objective directly to the base model (RL-Zero), without the SFT warm start, yields a worse trade-off, underscoring the role of Stage 1 in stabilizing long-horizon on-policy optimization. The final model is competitive with larger 3B–4B models, including a YaRN-based context-extension model from the same family, suggesting that the improvements come from the training recipe rather than scale alone. Appendix B.3 provides complementary generalization checks along two axes: family transfer to Gemma4-2B and same-family scaling to Qwen3-4B.

4.3 Dissecting the Training Recipe

Teacher-anchored RL stabilizes on-policy optimization and improves long-context reasoning. Figure 4 compares GRPO, GRPO with reference-policy KL regularization ($\beta_{\text{ref}} = 0.5$), OPD, and our teacher-anchored RL objective under the same SFT warm start, one-epoch RL budget, and rollout settings from Section 3. This matched-budget comparison isolates sparse outcome optimization, reference-policy anchoring, dense teacher guidance, and their combination. GRPO improves reward but exhibits a noisier reward trajectory, underscoring the difficulty of optimizing long-context behavior from sparse trajectory-level rewards. Adding reference-policy KL yields only modest improvements over GRPO, suggesting that the gains are not due to KL regularization alone, but to anchoring on-policy updates to a stronger teacher. OPD provides dense token-level supervision and smoother updates, but yields smaller long-context gains because it does not optimize task rewards. Our teacher-anchored objective combines outcome-reward optimization with dense teacher guidance, achieving the highest final reward among reward-optimizing methods, the strongest long-context performance, and unchanged short-context accuracy.

Teacher quality matters for teacher-anchored RL. Figure 5 compares our teacher-anchored RL stage using either a stronger external teacher or self-distillation from the current policy (Zhao et al., 2026). Self-distillation already improves over the baseline, showing that the teacher anchor need not come from a separate model. However, the stronger external teacher yields substantially larger gains, indicating that teacher-signal quality is an important factor in long-context reward optimization.

Tuning β: Regularization vs. Exploration. Figure 6 shows the effect of the teacher-anchoring strength β during RL. At $\beta = 0$, the objective reduces to GRPO and performs worst, highlighting

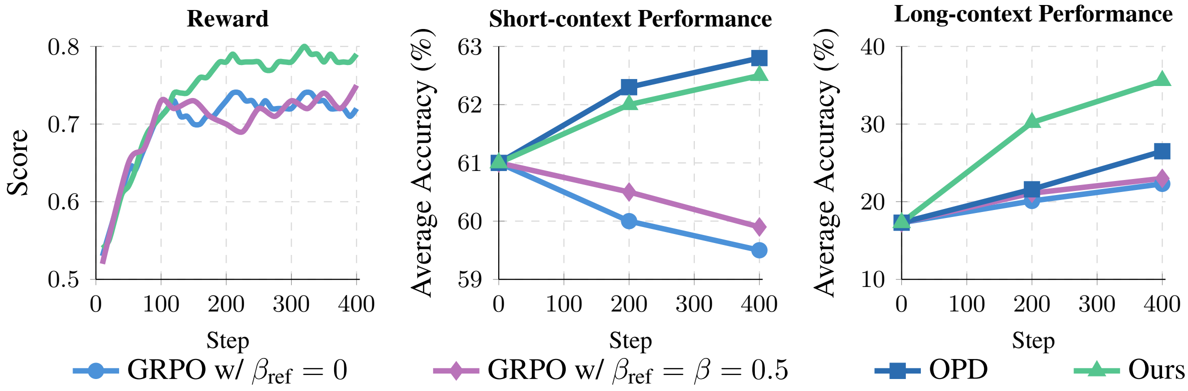

Figure 4: Comparison of GRPO with and without reference KL (Shao et al., 2024), OPD (Agarwal et al., 2024), and our teacher-anchored RL objective. OPD has no reward trajectory because it does not optimize rewards.

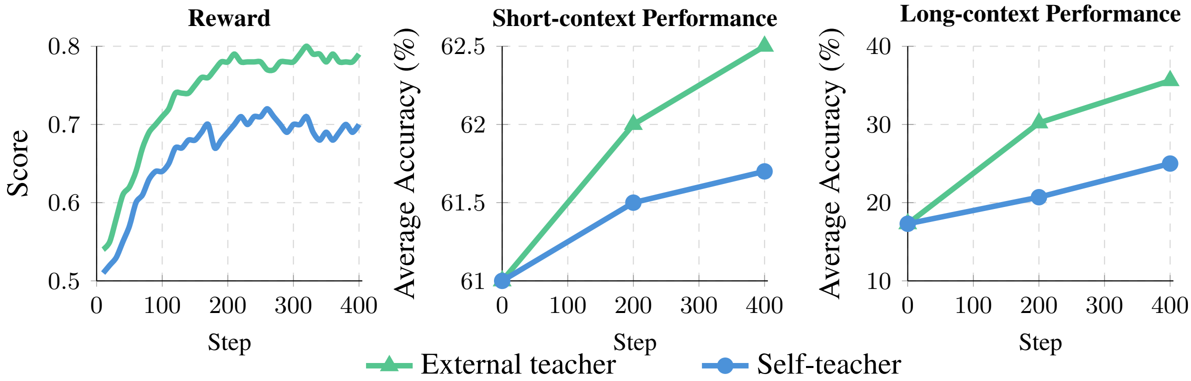

Figure 5: Comparison of external-teacher and self-teacher variants of our teacher-anchored RL objective, showing reward trajectories and short-/long-context evaluation performance.

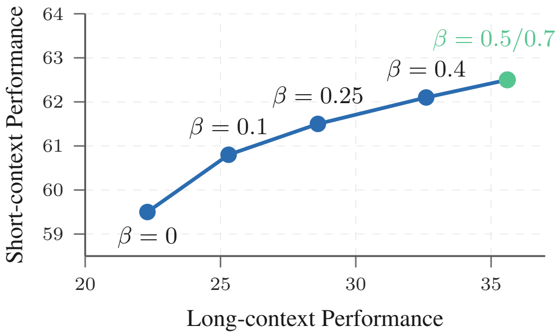

Figure 6: Effect of β during the RL stage on the short-/long-context performance trade-off.

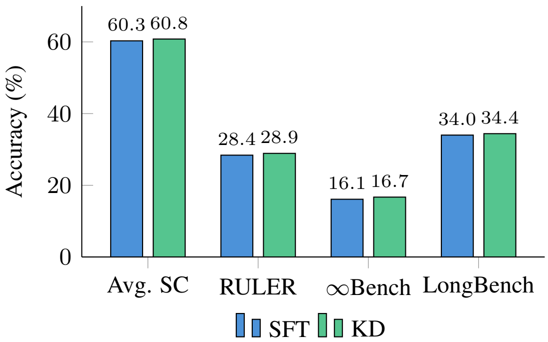

Figure 7: Comparison of SFT and KD on short-context (SC) vs. long-context benchmarks.

the difficulty of sparse-reward optimization without teacher guidance. Larger β improves both short- and long-context performance by regularizing model-generated trajectories. Gains peak at $\beta = 0.5$ and remain comparable at $\beta = 0.7$, suggesting that moderate anchoring is sufficient and that further increases yield diminishing returns. Overall, β controls the trade-off between reward-driven exploration and teacher-anchored stability, consistent with Agarwal et al. (2024).

Cold-start initialization: SFT vs. KD. Figure 7 compares SFT and KD as off-policy cold-start initialization strategies. The two approaches yield similar performance on both short- and long-context benchmarks, with KD providing only marginal improvements while requiring teacher inference when logits are not cached. We there-

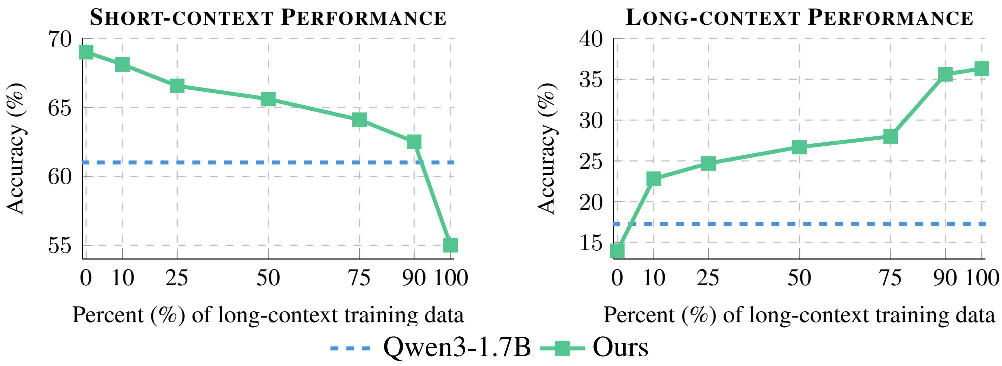

Figure 8: Accuracy on short- and long-context tasks across different short/long data mixes.

fore adopt SFT as the default initialization. We additionally use continued SFT from the resulting checkpoint as a matched off-policy-only baseline. This baseline uses the same 10/90 data mixture and context length, with a post-warm-up training budget comparable to that of the subsequent RL stage. Continued SFT leaves downstream performance essentially unchanged relative to the initial SFT checkpoint, indicating that the observed final improvements arise from the RL stage rather than from additional off-policy optimization.

Balancing short- and long-context data. We ablate the short-/long-context data mix under a fixed token budget. Figure 8 shows that short-context accuracy is robust to the mix: only 10% short-context data is enough to preserve performance. Long-context accuracy, however, drops as short-context data increases. We therefore use a 10/90 short-/long-context mix, which preserves short-context capabilities while maximizing long-context gains.

5 Related Work

Off-Policy vs. On-Policy Methods. Alignment methods for language models include off-policy methods, such as supervised fine-tuning, distillation, DPO (Rafailov et al., 2024), and KTO (Ethayarajh et al., 2024), which optimize fixed demonstrations or preferences. These methods are stable and compute-efficient, but can suffer from train–inference mismatch and exposure bias (Bengio et al., 2015; Ranzato et al., 2016). By contrast, on-policy methods optimize samples from the current policy, including PPO (Schulman et al., 2017), GRPO (Shao et al., 2024), DAPO (Yu et al., 2025), Dr. GRPO (Liu et al., 2025), and on-policy distillation methods (Agarwal et al., 2024; Yang et al., 2025; Zhao et al., 2026; Hübotter et al., 2026). This distinction is especially important for long-horizon generation, where local errors compound over extended rollouts. Our recipe combines these strengths through an off-policy cold start followed by teacher-anchored on-policy optimization.

Long-Context Adaptation and Post-Training. Prior work on long-context LLMs has focused either on extending context windows through architectural or positional modifications (Su et al., 2024; Chen et al., 2023; Peng et al., 2024), or on adapting models with static long-context data (Bai et al., 2024a; Chen et al., 2024). While effective, static supervision does not train models on their own long-horizon failure modes and is often centered on narrow formats such as retrieval or summarization, despite the need for robust distant-dependency reasoning (Liu et al., 2024; Bai et al., 2024b; Hsieh et al., 2024). Our approach addresses this by using broader supervision from LONGBLOCKS together with optimization on model-generated trajectories.

Dense Guidance for Long-Context Alignment. Recent work has begun to explore on-policy alignment for long-form generation and reasoning (Bai et al., 2024c; Zhang et al., 2025; Wan et al., 2025), showing that long-context post-training can benefit from reward-driven or preference-based optimization. Process-supervised reward models can densify sparse rewards (Lightman et al., 2024; Khalifa et al., 2025), but require reliable step-level supervision, which is scarce and difficult to define uniformly across retrieval, summarization, classification, and multilingual document QA. In parallel, RL–distillation and dense-guidance methods use token-level supervision to stabilize or complement on-policy learning (Agarwal et al., 2024; Yang et al., 2025; Hübotter et al., 2026; Zhao et al., 2026). However, their role in long-context alignment remains underexplored when sparse rewards, long rollouts, and short-context retention interact. We study this regime by pairing GRPO-style reward optimization with an on-policy teacher anchor, yielding a recipe for stable long-context alignment.

6 Conclusion

We studied how to align LLMs for long-context reasoning by combining off-policy warm-up, sparse-reward on-policy RL, and on-policy distillation. Our results show that these ingredients address complementary failure modes: SFT stabilizes long-horizon rollouts, GRPO adapts the model to its own generations, and teacher anchoring provides dense guidance for sparse-reward optimization.

We introduced a two-stage recipe, supported by LongBlocks, that improves long-context performance while preserving short-context capabilities. Ablations show that the gains come not from KL regularization alone, but from anchoring reward optimization to a stronger teacher. Overall, outcome-level rewards and dense teacher guidance offer a practical path toward stable long-context reasoning.

Limitations

Due to computational constraints, we prioritized component-wise informative ablation studies. Appendix B.3 provides initial evidence across one additional model family and one larger same-family student, but broader architecture and scale sweeps remain future work. Our ablations compare component-matched variants rather than exhaustively tuned alternatives such as preference optimization (e.g., DPO). We focus on teacher-anchored RL, which depends on teacher and reward quality, leaving aside self-distillation. However, at very long context lengths, on-policy training is costly due to multiple rollouts and teacher forward passes, motivating cached logits, selective distillation, or self-distillation. Finally, LongBlocks is synthetic and LLM-filtered and may inherit artifacts or biases from its sources, generator, or verifier. Although LongBlocks is task-family balanced, we leave fine-grained contribution analysis to future work.

Ethical Considerations

All source artifacts are used for research purposes in accordance with their licenses, terms of use, and access conditions. LongBlocks is released under CC BY-SA 4.0 where redistribution is permitted. For non-redistributable research content, we provide reconstruction instructions in the dataset card rather than redistributing the underlying documents. Although we use public or pre-filtered sources and apply cleaning, deduplication, and LLM-based verification, LongBlocks may still inherit artifacts, biases, offensive content, or personally identifying information from its sources, generator, or verifier. LongBlocks is intended for research on long-context post-training.

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A Multilingual Synthetic Data Generation for Long Context

A.1 Generation Prompts

We present the prompts used with `Qwen3-Next-80B-A3B-Thinking` to generate multilingual QA pairs. We discard incomplete, unsupported, weakly grounded, or locally answerable examples.

Summary Question Prompt

You are given the following document in `{language}`:

`{document}`

Your task is to create one **high-quality summary question** about the document and its corresponding answer.

Instructions:

- The question must require **synthesizing information from the entire document**.
- The answer must provide a **concise but complete summary** that captures all major themes, ideas, and relationships, reflecting a full and accurate understanding of the text.
- Your output must follow exactly the format below and should be in `{language}`.
- Do not include anything else.

Output Format:

```
Question: <question>
Answer: <answer>
```

Information-Retrieval Question Prompt

You are given the following document in `{language}`:

`{document}`

Your task is to create one **high-quality information-retrieval question** about the document and its corresponding answer.

Instructions:

- The question must require locating and **extracting information from multiple parts** of the document.
- Ensure that it can only be answered by someone who has **thoroughly read and understood the full text**.
- The answer must provide the **precise information requested**, directly grounded in the document.
- Your output must follow exactly the format below and should be in `{language}`.
- Do not include anything else.

Output Format:

```
Question: <question>
Answer: <answer>
```

Multi-Hop Reasoning Question Prompt

You are given the following document in `{language}`:

`{document}`

Your task is to create one **high-quality multi-hop reasoning question** about the document and its corresponding answer.

Instructions:

- The question must require **reasoning across multiple parts** of the document.
- It should combine information from **different sections, themes, or ideas**, requiring logical synthesis of the full text.
- The answer must clearly **explain the reasoning steps** or connections involved.
- Your output must follow exactly the format below and should be in `{language}`.
- Do not include anything else.

Output Format:

```
Question: <question>
Answer: <answer>
```

Comprehensive Question Prompt

You are given the following document in `{language}`:

`{document}`

Your task is to create one **high-quality question** and its corresponding answer.

Instructions:

- The question must require information from the **entire document**, not just a small part.
- It should involve **multiple aspects or themes** so that it can only be answered by someone who has fully read and understood the document.
- Your output must follow exactly the format below and should be in `{language}`.
- Do not include anything else.

Output Format:

```
Question: <question>
Answer: <answer>
```

Dataset	Scale	Max Context	Multilingual	Coverage	Supervision Focus
LongAlpaca	12,000	100K	× No	Long-document QA	Instruction following
LongAlign	9,888	64K	× No	Long-document QA	Instruction following
LongBlocks	193,219	128K	✓ Yes	Multilingual documents and code	Long-context reasoning

Table 3: Comparison between LongBlocks and representative long-context instruction datasets.

Classification Question Prompt

You are given the following document in `{language}`:

`{document}`

Your task is to create one **high-quality classification question** about the document and its corresponding answer.

Instructions:

- The question must require **identifying or inferring categories, labels, or sentiment** from content spread across the entire document.
- Ensure that the classification **cannot be made without reading and understanding the full text**.
- The answer must clearly provide the correct classification or sentiment label and show that it is grounded in the document's overall content.
- Your output must be in `{language}` and follow the format below exactly.
- Do not include anything else.

Output Format:

```
Question: <question>
Answer: <answer>
```

A.2 Dataset Composition

We balance LongBlocks uniformly across five task families: summary, multi-part information retrieval, multi-hop reasoning, comprehensive QA, and classification. Table 3 compares LongBlocks with two public long-context instruction datasets commonly used for long-context adaptation: LongAlpaca (Chen et al., 2024) and LongAlign (Bai et al., 2024a). LongBlocks provides reasoning-oriented long-context supervision at larger scale and with broader multilingual and source coverage. Its generation templates target complementary task families, including evidence aggregation, multi-hop reasoning, summarization, classification, and grounded long-form generation; for reasoning-focused examples, reference answers expose the relevant reasoning steps or cross-document connections. Before filtering, we sample the same number of source documents for each generation template, ensuring uniform task-family coverage at generation time. We also apply benchmark-overlap controls by canonicalizing each LongBlocks source document and each available benchmark context through lowercasing, markup removal, and whitespace normalization, then removing exact document-level matches and fuzzy near-duplicates identified through high-overlap character- and word-level fingerprints against RULER, OneRuler, LongBench, and ∞Bench.

Language-wise composition of LongBlocks. Figure 9 summarizes the language composition of the generated dataset. The dataset covers 31 natural languages, with contributions from heterogeneous sources including Wikipedia, Institutional Books, FineWeb2-HQ, ArXiv, PGBooks, and StackExchange. High-resource languages receive contributions from multiple sources, while Wikipedia broadens coverage to lower-resource languages.

A.3 Reward and Judge Reliability

LLM-as-a-Judge Prompt for Reward Computation. During RL, `Qwen3-4B` is used as the LLM judge to compute binary rewards by comparing student responses against the ground-truth answers.

Reward reliability. We evaluate the LM judge used for long-context rewards with human validation and large-scale judge agreement. We compare Qwen3-4B decisions with human annotations on a 250-example stratified set spanning the five LongBlocks task families and balanced across Qwen3-4B reward labels. We also compare Qwen3-4B with Qwen3-32B and Gemma-3-27B on balanced 10,000-example subsets with equal positive and negative decisions from the comparison judge, measuring agreement on both accepted and rejected cases. We report agreement, Cohen's κ, positive-label F1, and false-positive/false-negative rates. As shown in Table 4, Qwen3-4B shows strong agreement with human annotations and with both alternative judges, including the different-family Gemma-3-27B judge, suggesting that the binary reward signal is consistent across human and model-based evaluations.

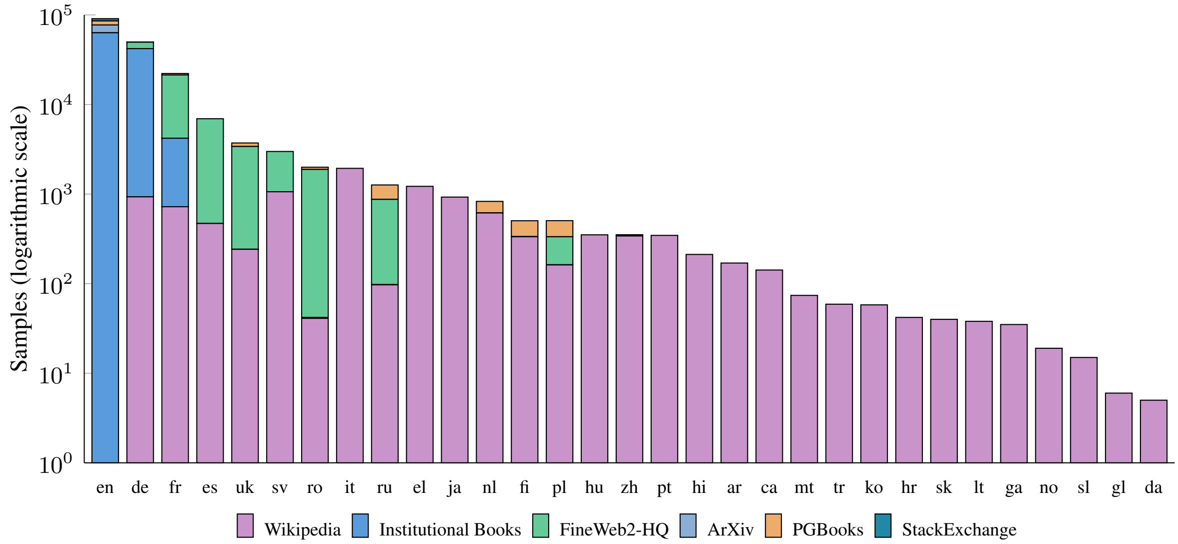

Figure 9: Language-wise composition of the corpora used to generate multilingual synthetic long-context data for LONGBLOCKS. Each stacked bar corresponds to one language, and colored segments denote the number of samples contributed by each source. Programming and markup languages are uniformly sampled from Stack-Edu but omitted for readability.

Comparison	**Examples**	**Agreement (%)**	κ	**Pos. F1**	**FPR/FNR (%)**
Qwen3-4B vs. Human	250	84.8	0.696	0.847	14.4 / 16.0
Qwen3-4B vs. Qwen3-32B	10,000	91.6	0.832	0.915	7.4 / 9.4
Qwen3-4B vs. Gemma-3-27B	10,000	87.0	0.740	0.866	10.0 / 15.9

Table 4: Reliability analysis of the Qwen3-4B judge used for binary long-context rewards. Human validation is performed on a 250-example stratified subset, while model–model agreement is measured on balanced 10,000-example subsets with equal positive and negative decisions from the comparison judge.

LLM Judge Verification Prompt

You will receive a **question**, a **response** generated by an LLM, and a **ground-truth answer**.
Your task is to determine whether the response is **accurate** according to the ground-truth.

Instructions:

- Respond with **only a single integer**:
  - **1** → The response is accurate and aligns with the ground-truth.
  - **0** → The response is inaccurate or does not match the ground-truth.
- Do not include any explanations, text, or commentary—just the integer.

Input Format:

```
Question: {question}
Response: {response}
Ground-truth: {ground-truth}
```

Output Format:

```
Answer: <int>
```

B Detailed Results

This appendix reports full benchmark score breakdowns, including run-to-run standard deviations, long-context results, and multilingual results.

B.1 Run-to-Run Variability

Evaluation with reasoning models can be computationally expensive and subject to nontrivial run-to-run variability under stochastic decoding. To account for this variability, we evaluate all main benchmark results over five runs using the shared decoding configuration described in Section 4.1. Main text reports the mean score across runs for readability, while this section reports the standard deviation across five runs for each benchmark. As shown in Table 5, these results indicate that our post-training approach does not introduce additional variability, with standard deviations that remain broadly in line with those of reasoning models, including Qwen3-1.7B.

Benchmark	Qwen3-1.7B	Ours
IFEval	0.6608	0.7155
IFBench	0.9901	0.6487
MMLU-Pro	0.1441	0.1427
GPQA ♦	1.9992	1.9499
GSM8K	0.4654	0.4461
MATH-500	0.4775	0.3578
HumanEval	0.8627	0.7550
HumanEval+	0.5561	0.4650
RULER	0.3030	0.1500
LongBench	0.2550	0.2168
∞Bench	0.3265	0.3918

Table 5: Standard deviation across five evaluation runs for the main benchmark suite.

B.2 Benchmark Breakdowns

Tables 6–11 provide the full multilingual and long-context benchmark breakdowns supporting the aggregate results in the main text.

Language	Qwen3-1.7B	Ours
ar	53.2	**54.1**
ca	**60.6**	60.5
da	**58.8**	58.0
de	61.1	**62.8**
es	62.7	**62.8**
fr	**63.0**	62.4
hi	49.7	**50.6**
hu	**56.2**	**56.2**
it	62.7	**62.9**
nl	**61.3**	61.0
pt	62.7	**63.5**
ro	**61.2**	60.5
ru	61.4	**61.6**
sv	**60.5**	60.3
uk	58.5	**60.2**
zh	55.4	**55.8**
Average	59.3	**59.6**

Table 6: MMMLU (OpenAI, 2024) results by language.

Language	Qwen3-1.7B	Ours
de	**80.8**	79.2
es	79.2	**80.0**
fr	**81.2**	80.8
ja	52.8	**57.6**
ru	**70.8**	**70.8**
zh	70.4	**71.2**
Average	72.5	**73.3**

Table 7: MGSM (Shi et al., 2022) results by language.

Category	Qwen3-1.7B	Ours
Single-Document QA	34.3	**39.2**
Multi-Document QA	23.4	**43.1**
Summarization	20.0	**20.2**
Few-shot Learning	42.9	**43.3**
Synthetic Tasks	**65.3**	65.0
Code Completion	20.3	**26.9**
Average	34.4	**39.6**

Table 8: LongBench (Bai et al., 2024b).

Category	Qwen3-1.7B	Ours
R.PassKey	13.9	**100.0**
R.Number	15.4	**93.2**
Retrieve.KV	**0.0**	**0.0**
En.Sum	11.5	**11.8**
En.QA	0.1	**8.1**
Zh.QA	3.9	**5.7**
En.MC	18.8	**30.1**
En.Dia	0.5	**1.2**
Code.Debug	3.7	**4.6**
Code.Run	0.5	**1.5**
Math.Calc	**6.0**	**6.0**
Math.Find	4.9	**10.6**
Average	6.6	**22.7**

Table 9: ∞Bench (Zhang et al., 2024), including all benchmark categories. Many low scores (e.g., *Retrieve.KV*, *En.QA*, *Code.Run*, *Math.Calc*) reflect the difficulty of reasoning over extremely long contexts.

Model	Length	Avg.	S1	S2	S3	MK1	MK2	MK3	MV	MQ	VT	CWE	FWE	SQA	HQA
Qwen3 1.7B	4K	87.4	100.0	100.0	99.8	100.0	95.4	98.6	99.1	100.0	99.0	91.8	75.7	42.0	34.2
	8K	84.2	100.0	100.0	99.6	100.0	92.4	92.6	97.1	99.8	97.3	73.9	76.3	32.2	34.0
	16K	81.0	100.0	100.0	100.0	99.4	81.8	93.8	93.2	99.6	92.5	46.4	88.7	29.5	28.6
	32K	72.0	100.0	100.0	99.8	89.6	74.6	45.2	89.0	96.5	88.8	19.4	79.9	28.0	25.6
	64K	51.5	100.0	97.2	90.6	55.0	3.2	0.2	64.2	84.3	55.2	7.5	61.6	27.8	22.3
	128K	11.0	61.0	3.0	3.6	7.4	0.0	0.0	0.0	1.7	0.4	0.2	58.6	3.7	3.6
Ours	4K	88.3	100.0	100.0	99.8	99.8	97.2	99.6	91.7	100.0	99.1	90.5	82.7	47.0	40.2
	8K	85.4	100.0	100.0	99.8	99.6	93.2	97.4	94.7	99.9	99.1	83.5	73.1	34.4	36.0
	16K	84.7	100.0	100.0	97.0	99.6	90.0	97.4	92.3	100.0	91.5	77.5	88.2	32.5	35.0
	32K	80.4	100.0	100.0	98.2	99.6	91.6	80.4	91.3	98.7	80.7	71.2	67.1	32.1	33.8
	64K	65.6	100.0	100.0	98.6	85.6	32.0	22.4	92.7	95.0	70.4	48.3	44.6	35.7	27.2
	128K	44.5	76.8	66.0	65.8	45.6	13.2	2.6	44.9	49.0	73.9	27.6	78.1	19.6	15.4

Table 10: RULER (Hsieh et al., 2024) breakdown across sequence lengths.

Language	Qwen3-1.7B					Ours				
	Avg.	8K	32K	64K	128K	Avg.	8K	32K	64K	128K
de	63.3	99.0	84.8	69.5	0.1	75.1	99.0	97.5	74.1	29.6
es	63.8	98.4	85.0	70.8	1.0	78.3	99.3	96.6	74.1	43.3
fr	63.4	98.9	85.4	68.8	0.5	76.8	99.0	96.5	71.6	40.1
it	62.8	98.9	84.5	67.0	0.8	75.3	99.0	97.1	71.1	34.1
ja	64.9	99.0	86.4	74.2	0.0	73.9	99.0	96.3	79.4	21.0
ko	83.8	98.9	86.4	74.9	7.9	86.5	98.8	96.4	77.9	72.9
nl	62.4	98.1	83.4	68.1	0.1	74.9	98.8	97.6	71.5	31.5
pt	63.9	98.4	85.1	71.6	0.5	77.8	98.9	97.1	76.0	39.1
ru	63.7	99.0	85.4	68.9	1.5	78.3	98.9	96.3	74.0	43.8
zh	64.7	98.5	86.4	74.0	0.0	74.1	98.6	94.8	78.8	24.0
Average	65.7	98.7	85.3	70.8	1.2	77.1	98.9	96.6	74.9	37.9

Table 11: OneRULER (Kim et al., 2025) results by language.

B.3 Model Family Transfer and Scaling

We further assess the generality of our recipe along two axes. The `google/gemma-4-E2B-it` experiment tests transfer to a different small model family, while the Qwen3-4B experiment tests whether the same recipe remains effective for a larger student from the original family. The Gemma4-2B transfer setting follows the same trend as the primary Qwen3-1.7B experiment: long-context performance improves while short-context performance is retained. RULER 128K improves from 69.1 to 79.0, ∞Bench improves from 15.6 to 30.7, and LongBench improves from 44.4 to 48.5. The Qwen3-4B scaling setting shows the same pattern at a larger student scale. RULER 128K improves from 65.8 to 78.5, ∞Bench improves from 13.0 to 26.6, and LongBench improves from 42.9 to 46.9, while short-context metrics are retained or improve on most benchmarks. Together, the Gemma4-2B and Qwen3-4B results suggest that the recipe is not narrowly tied to the Qwen3-1.7B setup.

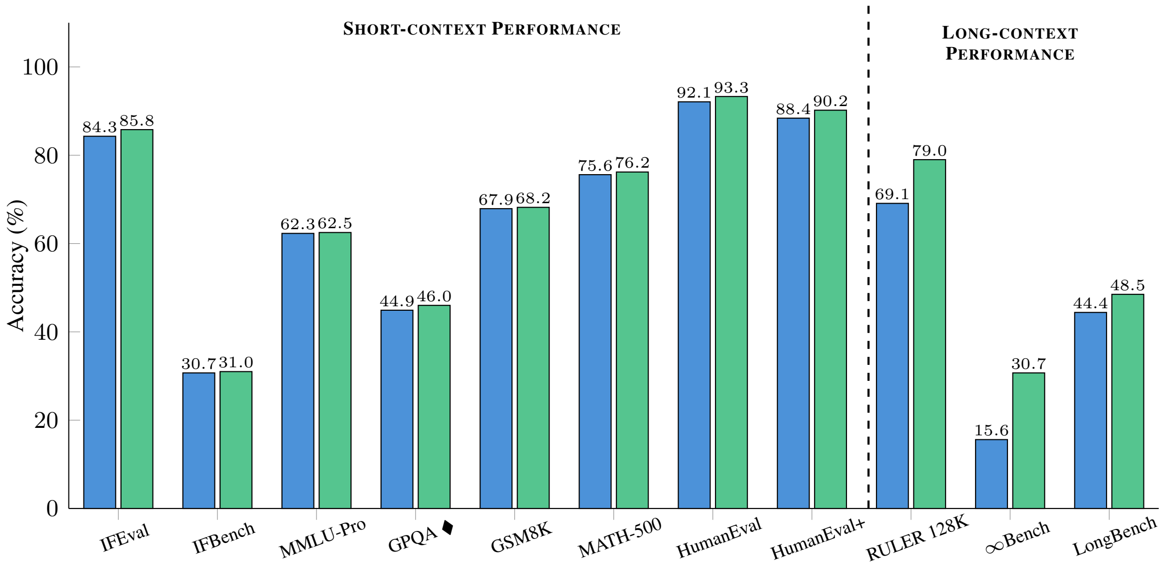

Figure 10: Gemma4-2B family-transfer benchmark performance.

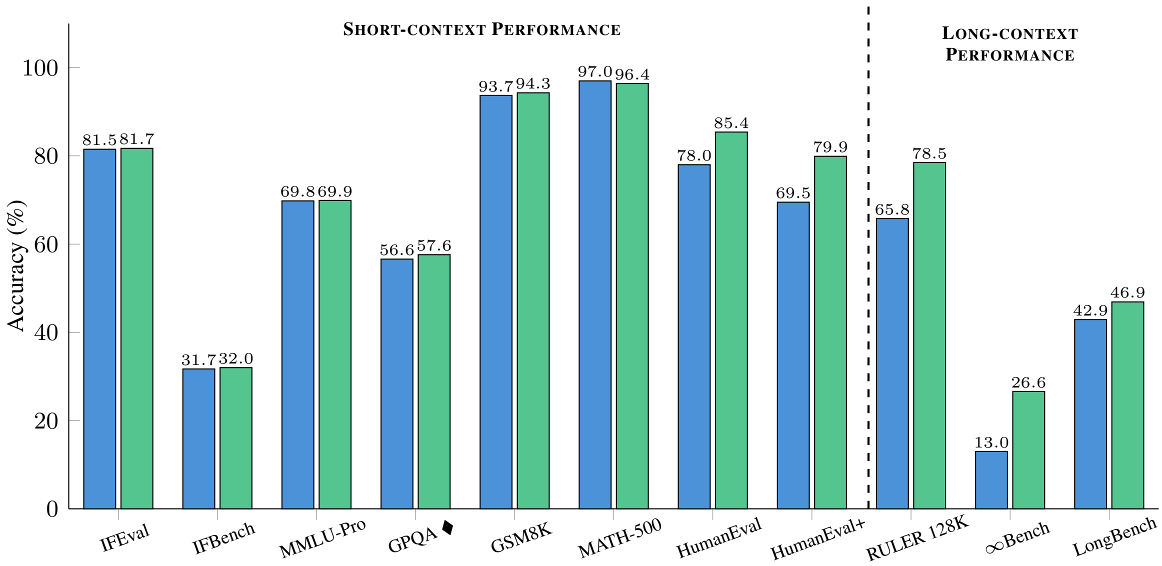

Figure 11: Qwen3-4B model-scaling benchmark performance.